

Forecasting Automotive Finance Trends: A Comparative Time Series Analysis



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Statistics 6313: Project

Executive Summary

Objective

This report analyzes the macroeconomic behavior of 60-month consumer auto loan rates to better understand how consumer borrowing costs shift over time. The primary goal is to evaluate predictive time series models that can forecast rate changes, ultimately providing actionable intelligence for strategic corporate finance and promotional retail planning.

Methodology

The analysis utilized quarterly commercial bank loan rate data sourced from the Federal Reserve Economic Data (FRED). Exploratory diagnostics confirmed the data exhibits non-stationary, wandering behavior driven by prolonged macroeconomic cycles rather than predictable seasonal retail trends. To account for this volatility, the data was difference to model the quarter-over-quarter velocity of rate changes. The study evaluated traditional statistical forecasting (ARIMA) against modern machine learning techniques (Multi-Layer Perceptron Neural Networks) and an ensemble approach.

Key Findings

- **Machine Learning Outperforms Traditional Statistics:** The Multi-Layer Perceptron (MLP) Neural Network was the superior predictive model. It achieved a lower Root Mean Square Error (RMSE) than the ARIMA(2,1,0) model, a finding validated through rigorous rolling window evaluations.
- **Absence of Retail Seasonality:** Statistical testing confirmed that macroeconomic auto loan rates do not exhibit annual seasonality. Rate fluctuations are driven by broader economic cycles and central bank policy rather than calendar months.
- **Univariate Model Limitations:** While the MLP model provided the best mathematical fit, univariate models inherently struggle to capture the full, complex volatility of the automotive finance market using historical loan rates alone.

Strategic Recommendations

To build upon this baseline analysis and create highly actionable business intelligence, future predictive modeling must incorporate exogenous economic variables. Integrating leading indicators like the Federal Funds Rate, the US Unemployment Rate, or vehicle inflation metrics via Vector Autoregression (VAR) will allow models to anticipate macroeconomic shocks before they fully price into consumer loans. Furthermore, pivoting the forecasting objective to predict directional rate movement (up or down) rather than precise percentage points will likely yield higher strategic value for risk management and advisory teams.

Exploratory Data Analysis and Stationarity

The dataset for this analysis, sourced from the Federal Reserve Economic Data ([FRED](#)), tracks the Finance Rate on Consumer Installment Loans at Commercial Banks (new Autos, 60-Month Loan). While FRED technically categorizes this series as a 'monthly' indicator, the underlying Federal Reserve survey is only conducted four times a year. To ensure mathematical integrity and avoid analyzing artificial gaps, this analysis treats the data as a continuous quarterly series, accurately reflecting the true macroeconomic reporting cycle for consumer automotive borrowing costs.

Initial visual inspection (Figure 1) of the realization confirms the data does not hover around a constant mean. This violates the core assumption of stationarity. The rate drops smoothly from 8% to just above 4% before climbing again, wandering wherever macroeconomic forces take it.

Our Sample Autocorrelations (ACF) plot agrees with this finding. The ACF starts at 1.0 and decays incredibly slowly, taking roughly 16 lags to cross the zero line. In a stationary series, this would drop to zero within the first two or three lags. Furthermore, the ACF plot shows no sign of cyclic pattern. If seasonal effects were present in this quarterly data, we would expect to see repeating bumps every 4 lags.

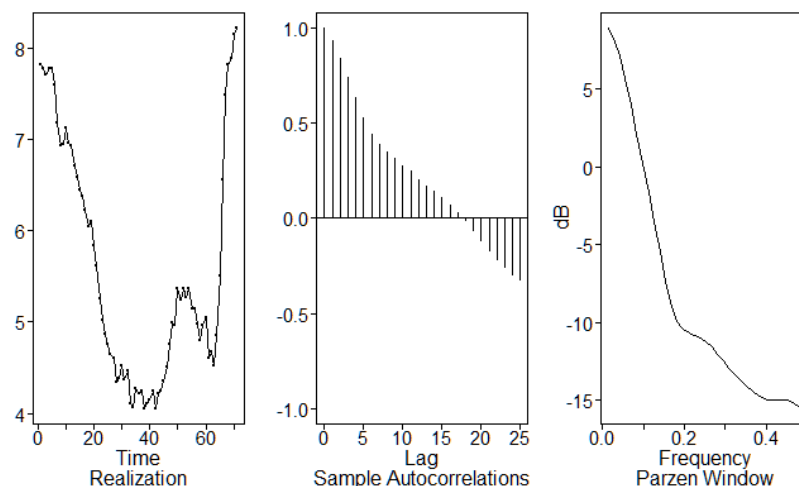


Figure 1: Realization, Sample Autocorrelations, & Spectral Density of Training Set

The Spectral Density plot reinforces this non-stationary behavior with a massive, dominant peak at $f = 0$, followed by an exponential drop in variance as frequency increases. A peak at $f = 0$ is the mathematical signature of a trend or wandering behavior. While there is a minor bump around $f = 0.23$, it is significantly weaker than the initial spike. This could hint at a faint seasonal pattern, but its comparative weakness suggests it may just be an artifact of Parzen Window smoothing. I do not recommend acting on this seasonal hint right now.

Overall, these visual findings make economic sense. Interest rates are heavily managed by the Federal Reserve and move in prolonged hiking or cutting cycles rather than bouncing randomly around a fixed historical average. Taking the first difference will clarify if the annual cycle is real or just noise.

Testing for Unit Roots

Because the series shows signs of wandering behavior, we must mathematically confirm the presence of a unit root to determine if we should fit an ARIMA model or a standard ARMA model. We evaluated this using the Augmented Dickey-Fuller (ADF) and KPSS tests.

The Augmented Dickey-Fuller test yielded a test statistic of -2.2454, which is greater than the 5% critical value of -2.89. This means we fail to reject the null hypothesis. There is not enough evidence to support that the data is stationary, indicating we should include a unit root in our model.

The KPSS test produced a test statistic of 1.0811 against a 5% critical value of 0.463. Because the hypothesis for KPSS is inverted compared to the ADF test, a test statistic greater than the critical value means we reject the null hypothesis of stationarity.

Both tests statistically confirm our visual investigation. The data is non-stationary and requires differencing.

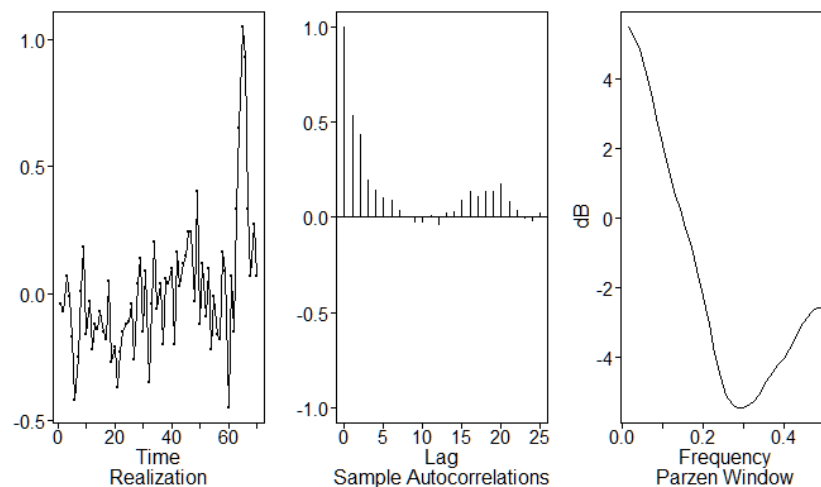


Figure 2: Realization, Sample Autocorrelations, & Spectral Density of Differenced Training Set

After taking the first difference, the realization now bounces around a constant mean of zero. (Figure 2) The ACF plot still does not show a trend for seasonality. If seasonality existed, we would see repeated spikes at lags 4, 8, and 12. Instead, lag 4 is lower than lag 3, and lag 8 is barely visible. Furthermore, the Spectral Density plot no longer shows the seasonal spike. This confirms the initial hint of an annual trend was simply noise.

Testing for a Deterministic Trend

With the data differenced, we tested for a deterministic trend using the Woodward-Bottone-Gray (WBG) Bootstrap Test. The WBG Test produced a p-value of 0.593, which is greater than our 0.05 alpha level. Therefore, we fail to reject the null hypothesis. There is not enough evidence to suggest a deterministic linear trend exists in the data.

Model Formulation and Evaluation

We selected an ARIMA(2,1,0) model, which relies on the previous two quarters to predict the next period. The p-value from the Ljung-Box Test is 0.998. Since this is greater than 0.05, we fail to reject the null hypothesis, confirming our residuals resemble white noise. This is the exact diagnostic result we want.

We also trained a Multi-Layer Perceptron (MLP) Neural Network. The neural network similarly determined that the change in loan rates from two quarters ago was primary driver for forecasting, completely ignoring the data from just one quarter prior.

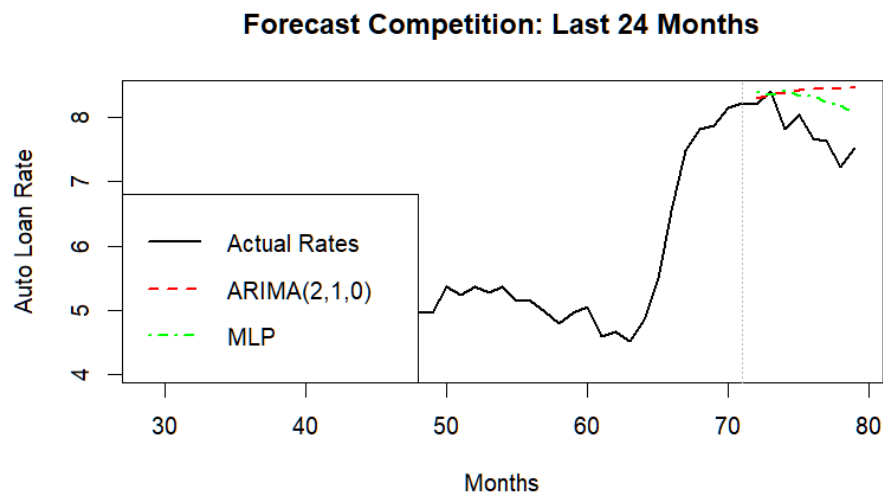


Figure 3: ARIMA & MLP predictions vs actual data

When comparing performance, the MLP Neural Network outperformed the ARIMA model. The ARIMA model produced a Root Mean Square Error (RMSE) of 0.721, while the MLP achieved a lower RMSE of 0.561. Visually, both models struggle to capture the true volatility of the test set. The ARIMA's poorer performance is largely due to its predictions following a flat, slightly upward trajectory, whereas the MLP correctly anticipated a downward slope. As statistician George Box famously noted, "All models are wrong, but some are useful." While neither model perfectly captures the raw behavior of auto loan rates, their baseline logic remains a helpful starting point.

Advanced Modeling

To improve predictive power, I developed an Ensemble model combining the ARIMA and MLP forecasts. This demonstrated how ensembling can be highly effective when combining two distinctly opposing models. While the Ensemble outperformed the standalone ARIMA, the isolated MLP Neural Network remained the superior model of the three.

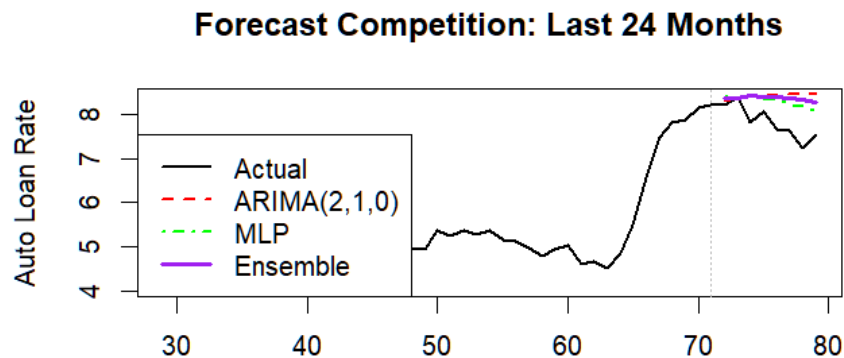


Figure 4: : ARIMA, MLP, & Ensemble predictions vs actual data

Next, I ran rolling window RMSE evaluations to test model stability over time.

Rolling Window RMSE Evaluation	
Model Performance Comparison (Lower is Better)	
Predictive Model	Rolling RMSE
ARIMA (2,1,0)	0.6110
MLP Neural Network	0.5763

Table 1: Rolling Window RMSE Model Evaluation

The rolling window approach revealed that both models are more robust than the initial static RMSE suggested. The ARIMA rolling window RMSE improved to 0.611, and the MLP rolling window RMSE improved to 0.576. This confirms our earlier finding that the MLP Neural Network is the strongest candidate.

Limitations & Future Considerations

Interest rates do not exist in a vacuum. They are heavily influenced by external economic forces. By strictly looking at historical loan rates, our current univariate models are blind to the broader macroeconomic environment.

To build highly effective models in the future, we should incorporate leading economic indicators such as the Federal Funds Rate, the US Unemployment Rate, or Total Vehicle Sales. We can use the Cross-Correlation Function to discover if a spike in one of these exogenous variables leads to a delayed shift in auto loan rates two quarters later. Additionally, we could build a Vector Autoregression (VAR) model to map the dynamic relationships between these datasets, allowing the model to adjust to sudden macroeconomic shocks that a simple ARIMA pipeline will always miss.

Finally, because auto loan rates are subject to high-frequency macroeconomic volatility, predicting the exact numerical rate for the next quarter is incredibly difficult. A highly practical alternative for corporate advisory would be to pivot this from a regression problem to a classification problem, focusing the model strictly on predicting the directional movement (will the rate rise or fall) rather than the precise percentage.

References

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